



2669 Martin Luther King Jr Blvd
Youngstown, OH 44510

VALLOUREC STAR LP
Seamless Tubular Products

Certified Test Report
07/11/2022
Revision: 0

Order: Y-221122445-001	Desc: Seamless Line Pipe- Normalized Rolled	Customer: Edgen Murray Corp.	SLN / (Heat): MG0363
Grade: API 5L X42/B N/BN PSL2 46th Edition May 01, 2019		Customer Order: DIS-284410	Split: 00
OD(in): 4.500	Wall(in): 0.337	Lbs/Ft: 15.00	Length: Double Random
End Finish: Plain End Beveled			
Comments: Melted and Manufactured in Youngstown, Ohio USA			
Pipe is also manufactured to: ASTM A106B/C-19a, ASTM A53B-20, ASME SA106B/C-2021, ASME SA53B-2021. Vallourec Star's QA program meets the requirements of DIN EN10204-2004 Type 3.1. Satisfies NACE MR0175/ISO15156:2015 and ANSI/NACE MR0103/ISO17945:2015.			
No repair welding has been conducted. Material is free from Mercury.			

Mechanical Properties

		Tensile Size Specimen Cross Section			Strength (ksi)			Elongation		
Orientation	Type	Width (in)	Thick (in)	Area (sqin)	Yield	Tensile	Y/T Ratio	Gage Length	% Elong	% RA
Longitudinal	Strip	1.000	0.350	0.3600	46.2	73.1	0.63	2	40.9	62.6

Required Min/Max Yield Strength (ksi): 42.1/71.8 Required Min/Max Tensile Strength (ksi): 70/95 Calculated Max Internal Yield (psi): 6051

Required Min Elongation(%): 30

Supplemental Requirements

Flattening Test	PASS
Hardness 3 Point	77.4 avg HRBW
Hydrostatic Pressure	4730 psi
Hydrostatic Hold Time	5 seconds

Chemical Analysis Chem Spec: 67

WT%	C	Mn	P	S	Si	Cu	Ni	Cr	Mo	Sn	Nb	V	Al	Ca	B	Ti	N	C.E.
Heat	0.18	0.83	0.010	0.003	0.22	0.21	0.07	0.10	0.02	0.010	0.000	0.001	0.034	0.0024	0.0001	0.003	0.0068	0.37
Product 1	0.18	0.83	0.011	0.003	0.22	0.22	0.07	0.09	0.02	0.009	0.000	0.001	0.033	0.0019	0.0001	0.003	0.0079	0.37
Product 2	0.18	0.82	0.009	0.003	0.22	0.22	0.07	0.09	0.02	0.009	0.000	0.001	0.034	0.0020	0.0001	0.003	0.0076	0.36

Charpy Impact Testing - Not required per API.

This material has been produced and tested in accordance with the requirements of applicable specifications unless otherwise listed below. We hereby certify that the above test results are representative of those contained in the records of the company. Any modification to this certification as provided by Vallourec Star without the expressed written consent of Vallourec Star negates the validity of this test report. Vallourec Star is not responsible for the inability of this material to meet specific applications.

THIS CERTIFICATE IS NOTARIZED ONLY WHEN REQUESTED.

Signed: 

Name: Jeff Schultz, Lab Supervisor

Date Approved: 07/11/2022

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Inspection Company: L Bay			

Non-Destructive Evaluation Criteria

EMI CHANNELS

ID

10% Longitudinal
10% Transverse

OD

5% Longitudinal
5% Transverse

ADDITIONAL REQUIREMENTS

Notch Length 1.5" Max
Notch Width .020" Max

FULL LENGTH WALL THICKNESS CHECK

87.5% Wall Reduction
Required Volumetric Coverage: 100%